Geospatial Analysis of Ideological Distribution and Campaign Finance in Washington State

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Introduction

The development of the Database on Ideology, Money in Politics, and Elections (DIME) and its Common-Space Campaign Finance Scores (CFscores) has provided researchers with a powerful tool to estimate the ideological positions of candidates, political committees, and donors using campaign contribution data (Bonica, 2013). Unlike roll-call vote measures which focus primarily on elected officials, campaign finance-based measures such as CFscores allow researchers to analyze not just the politicians, but the individuals, corporations, PACS that give to the politicians. These measures have contributed significantly to research on political polarization, representation, and donor influence in politics.

Within the literature, utilization of CFscores relies on non-spatial methods of analysis. Studies such as Bonica (2015) utilize CFscores to examine corporate influence in politics, demonstrating that corporate elites, unlike corporate PACs, exhibit more ideological diversity and a greater willingness to support non-incumbents. Similarly, Bonica, Rosenthal, and Rothman (2014) apply CFscores to analyze the political polarization of physicians, revealing shifting partisan alignment based on factors like gender, income, and medical specialty. Additionally, Bonica's (2015) study on state Supreme Court justices aggregates CFscores of the justices. Whilst the literature does not support use of CFscores in spatial analysis currently, there are other adjacent genres of literature that utilize spatial analysis in political science.

For example, Shi, Gong, and Wang (2021) apply spatial econometric models to examine how social networks and geographic proximity influence CEO compensation. Their findings reveal that CEOs in close geographic and social proximity tend to receive similar levels of compensation, particularly in firms with weaker corporate governance structures. This suggests that spatial relationships play a crucial role in shaping economic and political behaviors, highlighting the potential for incorporating spatial analysis into studies of campaign finance and ideological influence. Applying similar spatial techniques to CFscore data could provide new insights into how donor networks, geographic clustering, and social ties influence ideological alignment and financial support patterns in political campaigns.

While previous studies have provided valuable insights into group behaviors, they have primarily focused on ideology without considering its spatial dimension. This research project aims to bridge that gap by exploring CFscore data with geographic information systems (GIS), specifically ArcGIS, to analyze the geospatial distribution of ideology at the state level. Unlike prior research that emphasizes individual ideological placement, this project will examine spatial patterns of ideology within Washington State, identifying regional hotspots.

In addition to mapping ideological hotspots, this research will investigate two distinct donor groups: those who donate only once or twice during an election and those who contribute multiple times within the same election cycle. Many studies dismiss one-time or infrequent donors, arguing that a small number of contributions does not serve as a strong ideological proxy. However, this perspective overlooks the substantial financial impact these donors have on elections. By analyzing the spatial distribution of both donor types, this project will assess whether their spatial patterns align or diverge.

This research is largely exploratory, seeking to provide a new perspective on the relationship between ideology, campaign contributions, and geography. By shedding light on the spatial patterns of different donor behaviors, this project aims to inspire future scholars to expand on this framework and further explore the intersection of political geography and ideology.

Laymen Explanation of CFscores

To illustrate how CFscores function, consider a mother planning her son’s birthday party. Reflecting on a previous party he attended, she remembers her son's preference for pizza and cake, as he avoided vegetables. She also observed distinct eating habits among other children—some favored chicken nuggets, others preferred fruits and vegetables, while a few sampled various foods. In this analogy, her son and the other children represent "donors," each revealing their "preferences" through their food choices. The previous birthday child, who provided the food selection, symbolizes a political candidate offering ideological positions to attract support. The mother, analogous to an analyst, infers her son's food preferences from his previous choices, mirroring how CFscores infer donor ideologies based on their contribution patterns.

Consequently, the mother confidently predicts her son's preferences for pizza and cake at his upcoming party. Moreover, having mapped the other children's food choices from prior events, she accurately anticipates their preferences, thereby avoiding the need for direct consultation with each parent. Similarly, CFscores utilize indirect revealed-preference methods to estimate donor ideologies efficiently. Directly surveying donors is impractical, costly, and ethically challenging; thus, CFscores provide a viable alternative by indirectly revealing ideological preferences.

If her son primarily consumed foods offered at the last party, the mother infers the hosting child's preferences align with those foods, paralleling CFscores' assumption that politicians share ideologies with their donors. Mapping multiple children's preferences across various events creates a model of food preferences analogous to CFscores' creation of a common ideological space, facilitating comparisons across elections and candidates.

CFscores rely on extensive datasets compiled from several authoritative sources, including the Federal Election Commission (FEC), the Center for Responsive Politics (CRP), and the National Institute for Money in State Politics (NIMSP) (Bonica, 2013a). These institutions aggregate publicly available campaign contribution records, providing detailed insights into political financial transactions across numerous election cycles. Given the millions of contributions from individual donors and political organizations, data processing demands meticulous cleaning and standardization. Donors lack unique identifiers in public records, necessitating identity-resolution algorithms to match donor records based on names, addresses, occupations, and employers (Bonica, 2013b). Accurate donor tracking over time significantly enhances ideological estimations' reliability.

Data Handling

For the research project, data was downloaded from Adam Boncia's public DIME data page that is hosted by Stanford. This comprehensive dataset contains detailed itemized records of campaign contributions across multiple election cycles, from 2006 through 2024. As political donation activity has notably increased over time, the size of these files has similarly expanded, with compressed file sizes growing from approximately 2 GB in 2006 to roughly 17 GB by 2024.

Each record in the dataset includes numerous variables providing in-depth information. These variables include, but are not limited to, election cycle year, transaction ID, transaction type as coded by the Federal Election Commission (FEC), contribution amount, transaction date, unique contributor identification (bonica.cid), and detailed contributor demographics such as name, gender, occupation, employer, and address. Additionally, details such as name, party affiliation, recipient unique identifier, election type, and geographic coordinates of donator (latitude, longitude, census tract, and congressional district mappings) are provided. Importantly, the dataset also contains ideological positioning metrics known as CFscores for both contributors and recipients. Comprehensive documentation for each of these variables is available in the official DIME database guide (Bonica, 2024).

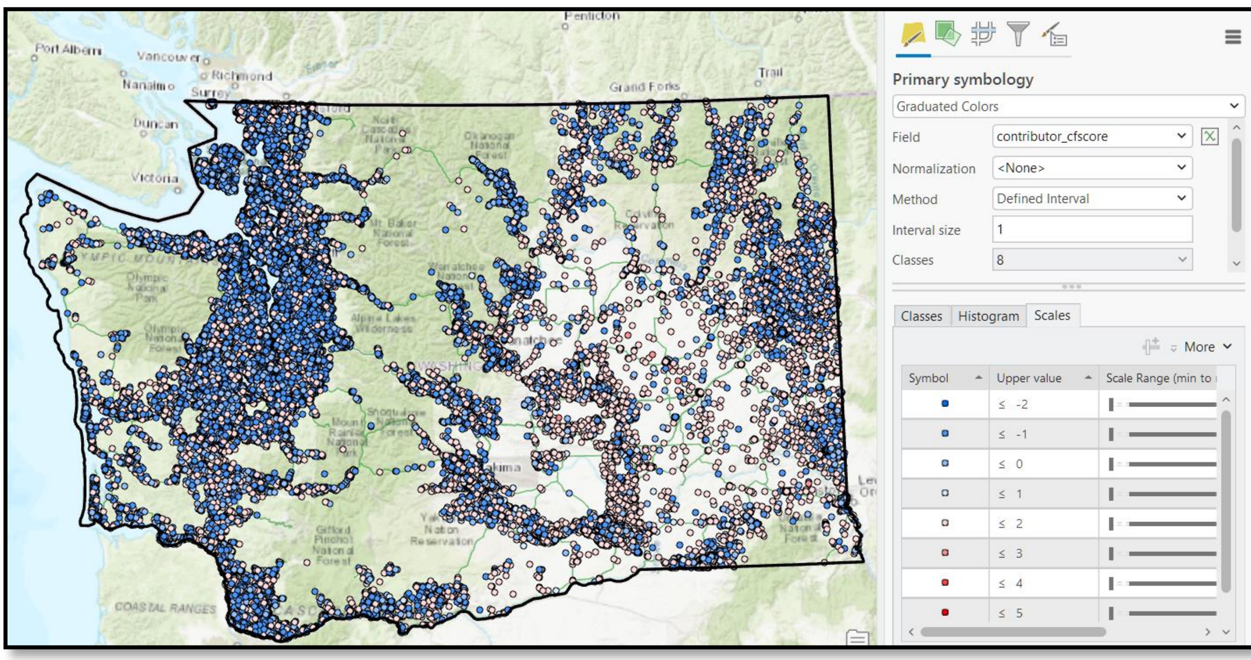
Given the substantial volume of data, reaching nearly 100 GB when uncompressed, conventional analytical methods quickly encounter hardware limitations, primarily related to insufficient Random Access Memory (RAM). These constraints make it challenging to execute queries and perform complex data analyses on standard computational resources. To address these limitations and enhance efficiency, the original compressed CSV files were converted into Parquet format using DuckDB. Parquet files significantly improve query performance, making them particularly suitable for managing and analyzing large-scale datasets.

DuckDB is advantageous because it can handle large-scale data queries even when computer hardware resources, such as RAM, are limited. It accomplishes this by temporarily using hard drive space (known as "disk spilling") when data is too large for memory, allowing operations to proceed smoothly without requiring significant hardware upgrades. DuckDB also employs external hashing, enabling complex data operations—such as joining multiple large tables, aggregating data, and generating summary statistics—even when data exceeds available memory.

To prepare and manage the data effectively, specific handling methods were applied. First, I queried observations specifically for Washington State where donations were made by individuals and not corporations. Next, I had to tackle instances where donator info changes during an election cycle. For instance, someone who previously donated in the 2010 election cycle might move before the cycle ends and donate again from the new address. This analysis presumes a person's most recent donation during an election cycle represent their residential location in that election cycle. Furthermore, the number of unique recipients supported by each contributor was counted. CFscores rely on whom people give to calculate ideology, some research posits its best practice to eliminate people who donate only once or twice during an election. Whilst this project is seeking to test that assumption, it is seeking to explore how geo-spatially related multi-time donators and one-off donators are. People that donate to one or two politicians are placed into group 1. People that donated three or more politicians are placed into group 2. This systematic handling and summarization approach allows for stronger confidence in the results but discounts the value that one- or two-time donators bring to election cycles.

Maps

Upon importation of the DIME data into ArcGIS. I instantly noticed the density of the data might be a challenge to work with. With so many observations through so many years, I suspected there would be a lot of noise in any attempt to derive insights from the data. In this following representation of the data, I used a simple graduated color scheme to depict liberal donators as blue and conservatives as red. It's obvious at first glance that some broad clustering is at work. Urban areas appear to have denser clusters of liberal donators, whereas rural regions have clusters of conservative donators. This is of no surprise to anyone who knows about politics and geography.



This research project focuses on the use of data to examine geospatial differences in the residences of Groups 1 and 2. While the next section will outline the methods used for data analysis, it is important to first acknowledge some limitations. One key issue is the accuracy of geocoding. Bonica Adam, in his write-up on this dataset, claimed that the data was geocoded with high accuracy, a statement that is somewhat fair. However, given the complexity of geocoding millions of home addresses across diverse geographic regions and governmental jurisdictions, achieving consistently high accuracy is challenging. Errors can arise not only from geocoding itself but also from inaccuracies in how individuals record their addresses when filing donation records.

Although this is not a criticism of the geocoding process per se, a significant number of data points are ultimately misclassified. For example, as shown in the following screenshot, many individuals appear to reside on sidewalks or streets, an obvious inaccuracy. This highlights the need for future analyses using this dataset at the household level to either improve geocoding precision or exclude improperly geocoded data.

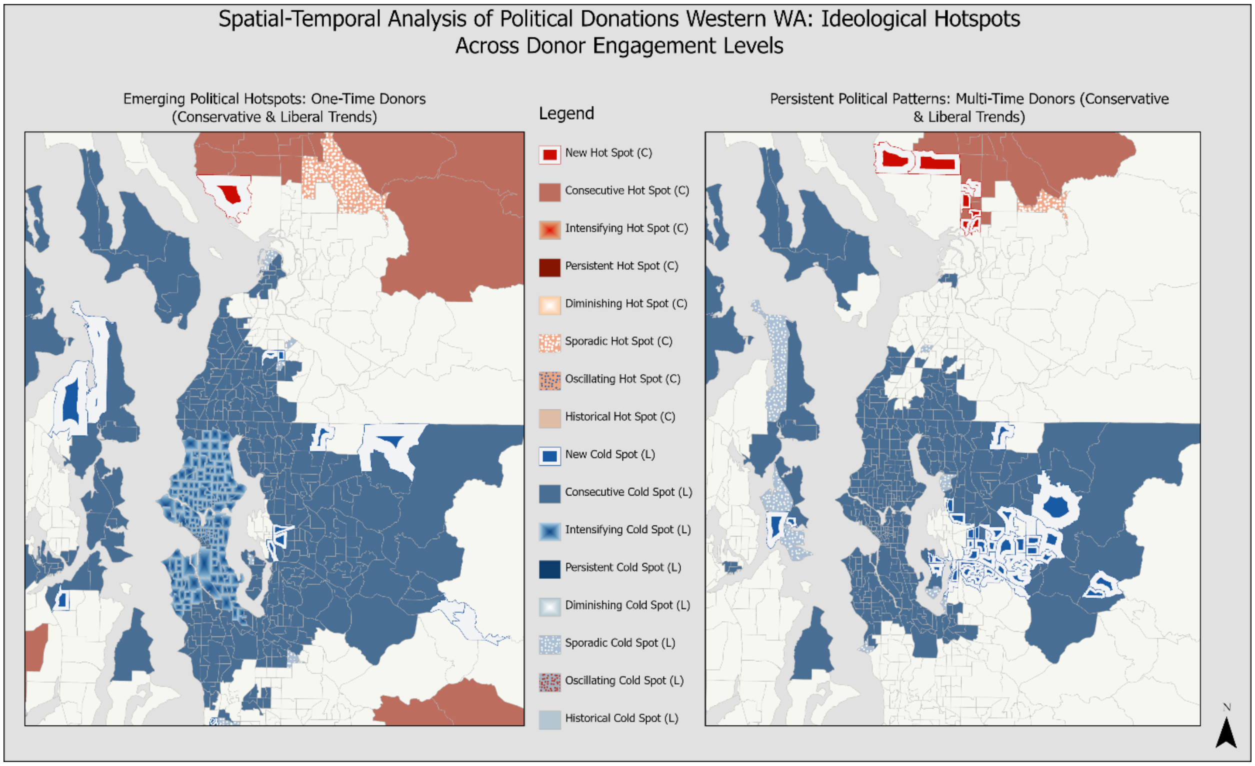
Emerging Hot Spot Analysis

ArcGIS offers a suite of tools that are invaluable for those exploring spatial data. The DIME dataset provided me with the advantage of time-stamped records spanning at least ten election cycles. This wealth of data allowed me to test and apply the Emerging Hot Spot Analysis tool. However, before utilizing this tool, I first needed to create space-time cubes.

Creating a space-time cube posed some minor challenges. The first decision was choosing the type of cube to use. While the term cube might suggest that a square serves as the standard two-dimensional time slice, multiple shape options are available. For instance, cubes can be based on hexagons, triangles, or even custom shapes such as census tracts or counties. For my analysis, I chose to aggregate ideological points at the census tract level.

The rationale behind this decision is that human communities are often structured around administrative boundaries. Using census tracts also provides a built-in urban-rural control when determining the size and shape of the space-time cube. Urban census tracts tend to be smaller than rural ones, which helps account for population density differences. While this approach is not perfect, it is sufficient for the level of analysis I am conducting. Finally, I delimited the data by one-off and multi-time donators. This resulted in two space time cube files.

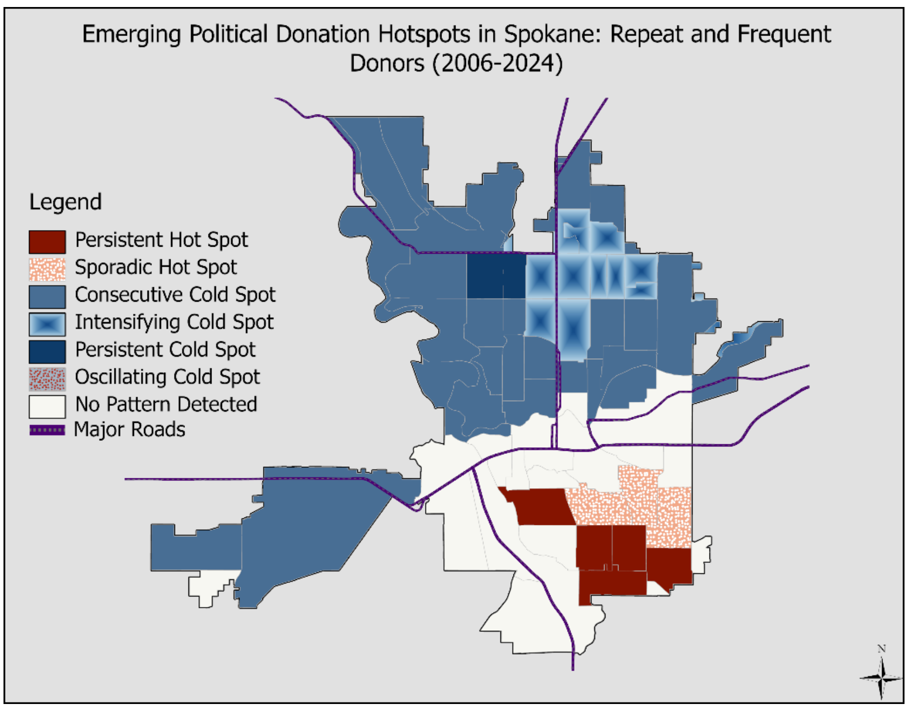
After creating the space-time cube files, I processed them using the Emerging Hot Spot Analysis tool. A key challenge in this analysis is balancing the detection of highly localized variations (using a small number of neighbors) with capturing broader spatial trends (using a larger neighbor count). Through testing, I found that using fewer than 10 neighbors resulted in overly fragmented patterns, potentially missing regional clustering effects. Conversely, setting the neighbor count too high (e.g., 20 or more) diluted localized trends, making them harder to identify. A setting of 10 neighbors struck the right balance, effectively highlighting local hot spots while preserving spatial dependence at a scale relevant to western Washington's census tracts. For the neighborhood time step, I selected three years—equivalent to six years in the context of biennial election cycles. This decision was based on the assumption that while individuals may struggle with long-term recall, they are more likely to remember trends over shorter periods. As a result, I structured the analysis around a moving window of three election cycles.

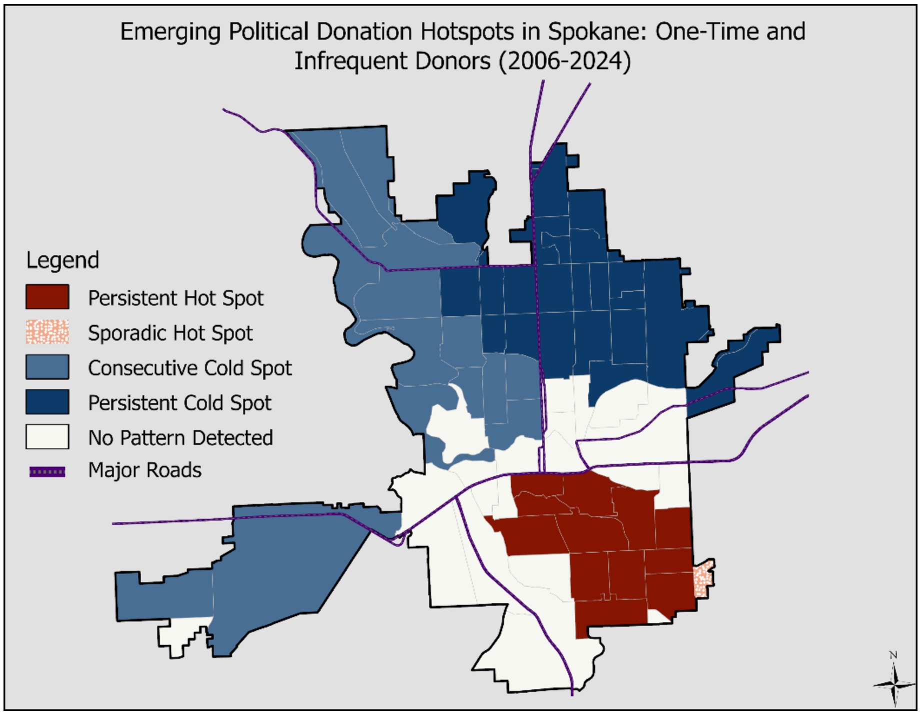


The final output consists of two maps illustrating ideological hot spots among liberal and conservative donors, further categorized by their donation frequency—either one-time or recurring. The maps reveal that one-time donors in Western Washington extend consistently into the outer suburbs of urban cores, whereas recurring donors tend to cluster closer to these central areas. Notably, recurring donations among liberals are growing in what is commonly known as the I-405 and I-90 corridors—areas characterized by new developments and high-income households. While this pattern is apparent to me based on my experience living in these regions, I aim to further explore the relationship between donor behavior and income by focusing on a specific city for deeper analysis.

For my next example, I focused on the city of Spokane. In this section, I examined which census tracts housed one-time and repeat donors, while also exploring how income levels might help explain the spatial distribution of these groups. Unlike the previous analysis, I did not differentiate donors by ideology. Whereas in the earlier maps red and blue were associated with political affiliation, in this case, red indicates an upward trend in donations, while blue represents a downward trend.

To conduct this analysis, I revisited the donation dataset, created two new space-time cube files, and reran the Emerging Hot Spot Analysis tool. For this iteration, I used the K-Nearest Neighbors method with 10 neighbors and a four-period time step, applying a neighborhood time step window to capture overarching trends while maintaining a localized focus. Additionally, I expanded the analysis to include all census tracts in Washington State before clipping the results to the Spokane city boundary using municipal boundary data.

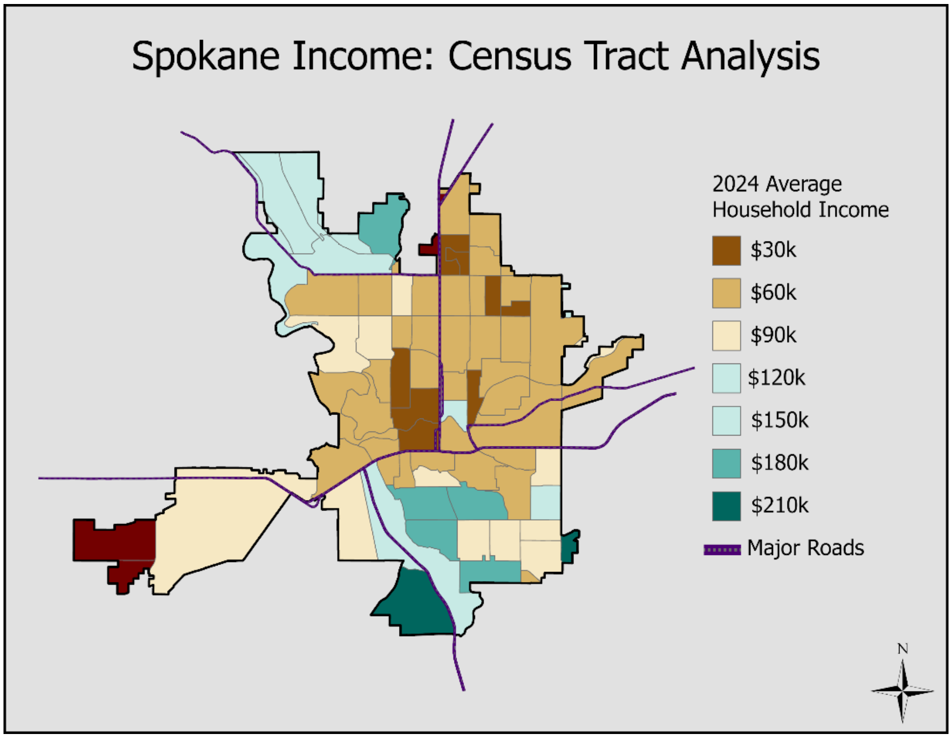




Both maps display emerging political donation hotspots in Spokane from 2006 to 2024, specifically focusing on repeat and frequent donors. Notably, both one-off and multi-time donators are clustering into the bottom right part of Spokane. In part, this trend may be influenced by demographic and economic factors. The bottom-right section of Spokane, where donation hotspots are emerging, coincides with areas experiencing higher-income.

Several possible explanations exist for this clustering:

1. **Affluence and Political Engagement** – Higher-income households tend to contribute more frequently to political causes, which may explain why repeat donors are concentrated in this region.
2. **Urban Development and Population Shifts** – This part of Spokane may be seeing an influx of newer, wealthier residents, increasing political donation activity.



Conversely, the northern and western sections of Spokane, which appear as persistent cold spots, generally align with lower-income areas, reinforcing the idea that financial capacity plays a role in donation frequency. Although, the northwest region of Spokane is of great affluence, but is expressed as a cold spot in both emerging hot spot analysis. There is possibility for that in the future the method used for this research should be revised. There is little knowledge on the internet related to this subject matter. The integration of CFscores with spatial analysis in this research project provides a novel perspective on ideological distribution and campaign finance behaviors. By leveraging the DIME dataset and geospatial techniques within ArcGIS, this project has successfully mapped ideological clustering across parts of Washington State, highlighting key differences between one-time and recurring donors.

The application of Emerging Hot Spot Analysis has uncovered meaningful trends in donation patterns over multiple election cycles. Recurring donors tend to cluster in more affluent and politically engaged regions, whereas one-time donors exhibit broader geographic distribution, particularly in suburban and developing areas. Despite these insights, this project also underscores the limitations and challenges associated with geocoding campaign finance data. While Bonica's CFscores offer a valuable measure of ideological placement, the accuracy of donor location data remains an ongoing concern.

References

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Appendix A

Python/DUCKDB Code

```
import duckdb
import os
import time
import pandas as pd
import geopandas as gpd
from shapely.geometry import Point

    # Define file paths for input and output data
    parquet_dir = r'D:\Contributor_Data'
    output_dir = r'D:\Contributor_Data\Output'
    temp_dir = r'D:\Contributor_Data\DuckDB_Temp'
    final_csv_path = os.path.join(output_dir, 'wa_contributions_all_years.csv')

    # Path to Washington state shapefile
    wa_shapefile_path = r'D:\Contributor_Data\State Boundry\WA_State_Boundary.shp'

    # List to store processed data for each biennial cycle
    all_years_data = []

    # Loop through each biennial cycle and process the data
    for year in range(2006, 2026, 2):
        parquet_file = os.path.join(parquet_dir, f'contribDB_{year}.parquet')
        duckdb_file = os.path.join(output_dir, f'wa_contributions_{year}.duckdb')

        if not os.path.exists(parquet_file):
            continue

        year_start_time = time.time()

        # Initialize DuckDB connection with necessary settings
        con = duckdb.connect(duckdb_file)
        con.execute("PRAGMA threads=6;")
        con.execute(f"SET temp_directory = '{temp_dir}';")
        con.execute("SET preserve_insertion_order = false;")
        con.execute("PRAGMA memory_limit='60GB';")

        # Begin transaction for database operations
        con.execute("BEGIN TRANSACTION;")
        con.execute(f"DROP TABLE IF EXISTS wa_contributions_{year};")

        # Create a temporary table with filtered contribution data
        con.execute(f"""
```

```
    CREATE TEMPORARY TABLE temp_wa_contributions AS
    SELECT
        LOWER("bonica.cid") AS bonica_cid,
        "bonica.rid" AS bonica_rid,
        "contributor.type" AS contrib_type,
        "contributor.state" AS contrib_state,
        "contributor.gender" AS gender,
        "contributor.employer" AS employer,
        "contributor.name" AS fullname,
        "contributor.address" AS address,
        "contributor.city" AS city,
        "contributor.zipcode" AS zip,
        CAST("amount" AS DOUBLE) AS amount,
        "seat" AS seat,
        "recipient.state" AS recipient_state,
        "latitude",
        "longitude",
        "cycle" AS original_cycle,
        "date",
        "is.corp" AS is_corp,
        "contributor.cfscore" AS contributor_cfscore
    FROM read_parquet('{parquet_file}')
    WHERE "contributor.state" = 'WA'
    AND "contributor.type" = 'I'
    AND "latitude" IS NOT NULL
    AND "longitude" IS NOT NULL
    AND CAST("amount" AS DOUBLE) > 0
""")

# Create a final table with aggregated data for each contributor
con.execute(f"""
    CREATE TABLE wa_contributions_{year} AS
    WITH
    latest_locations AS (
        SELECT
            bonica_cid,
            ROW_NUMBER() OVER (
                PARTITION BY bonica_cid
                ORDER BY "date" DESC
            ) AS row_num,
            latitude,
            longitude,
            "date" AS latest_date,
            address,
            city,
            zip
```

```
    FROM temp_wa_contributions
),
latest_locs_filtered AS (
    SELECT *
    FROM latest_locations
    WHERE row_num = 1
),
recipient_counts AS (
    SELECT
        bonica_cid,
        COUNT(DISTINCT bonica_rid) AS unique_recipient_count
    FROM temp_wa_contributions
    GROUP BY bonica_cid
),
amount_totals AS (
    SELECT
        bonica_cid,
        SUM(amount) AS total_amount
    FROM temp_wa_contributions
    GROUP BY bonica_cid
)
SELECT
    t.bonica_cid,
    l.latitude,
    l.longitude,
    l.latest_date,
    l.address,
    l.city,
    l.zip,
    a.total_amount,
    r.unique_recipient_count,
    MAX(t.gender) AS gender,
    MAX(t.employer) AS employer,
    MAX(t.fullname) AS fullname,
    MAX(t.contrib_state) AS state,
    CAST('{year}-01-01' AS DATE) AS cycle_date,
    MAX(t.original_cycle) AS original_cycle,
    CAST('{year}' AS VARCHAR) AS cycle,
    MAX(t.is_corp) AS is_corp,
    MAX(t.contributor_cfscore) AS contributor_cfscore
FROM temp_wa_contributions t
JOIN latest_locs_filtered l ON t.bonica_cid = l.bonica_cid
JOIN recipient_counts r ON t.bonica_cid = r.bonica_cid
JOIN amount_totals a ON t.bonica_cid = a.bonica_cid
GROUP BY
    t.bonica_cid,
```

```
        l.latitude,
        l.longitude,
        l.latest_date,
        l.address,
        l.city,
        l.zip,
        a.total_amount,
        r.unique_recipient_count
    """)

    # Retrieve processed data as a DataFrame
    year_data = con.execute(f"SELECT * FROM wa_contributions_{year}").fetchdf()
    all_years_data.append(year_data)

    # Cleanup temporary data and commit transaction
    con.execute("DROP TABLE temp_wa_contributions;")
    con.execute("COMMIT;")
    con.close()

# Combine all years' processed data into a single DataFrame
if all_years_data:
    combined_data = pd.concat(all_years_data, ignore_index=True)

    # Remove records with non-positive amounts
    combined_data = combined_data[combined_data['total_amount'] > 0]

    # Winsorize at the 99th percentile to cap extreme values
    percentile_99 = combined_data['total_amount'].quantile(0.99)
    combined_data.loc[combined_data['total_amount'] > percentile_99, 'total_amount'] =
percentile_99

    # Load Washington state boundary data for spatial filtering
    try:
        wa_shape = gpd.read_file(wa_shapefile_path)
        geometry = [Point(xy) for xy in zip(combined_data['longitude'],
combined_data['latitude'])]
        gdf = gpd.GeoDataFrame(combined_data, geometry=geometry, crs="EPSG:4326")
        wa_boundary = wa_shape.unary_union
        gdf['in_washington'] = gdf.geometry.within(wa_boundary)
        filtered_gdf = gdf[gdf['in_washington']]
        filtered_data = filtered_gdf.drop(columns=['geometry', 'in_washington'])
        filtered_data.to_csv(final_csv_path, index=False)
    except:
        combined_data.to_csv(final_csv_path, index=False)
```